AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *[GEN26-A0H-I055]*

Team Name: *[CLOVER]*

Domain: *[Domain 3]*

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Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

[Our device is designed to address the long wait for an early indication of physical changes in a finished lotion during stability testing. Problem statement: Conventional stability studies can take months, and samples are often checked only at scheduled intervals. our device aims to monitor a lotion under controlled temperature conditions and use infrared readings to flag specific optical changes—such as cloudiness or separation—so potential instability can be noticed sooner]

Problem Understanding

- [Who experiences this problem?]
  ANS: Cosmetic manufacturers and skincare brands
  Research and Development (RD) teams
  Cosmetic formulation scientists

- [What causes the problem?]
  1. These cosmetic products have formulations which contains the ingredients whose interactions can be complex 2. This laboratory method requires time and physical sample to validate stability 3. We have limited ability to predict the long -term behaviour quickly

- [How does it affect users or stakeholders?]
  1. Product launches will be delayed 2. The company will not be able to meet up to the market demand 3. Efficiency in cosmetic product development will be reduced 4. More material and chemical wastages will happen.

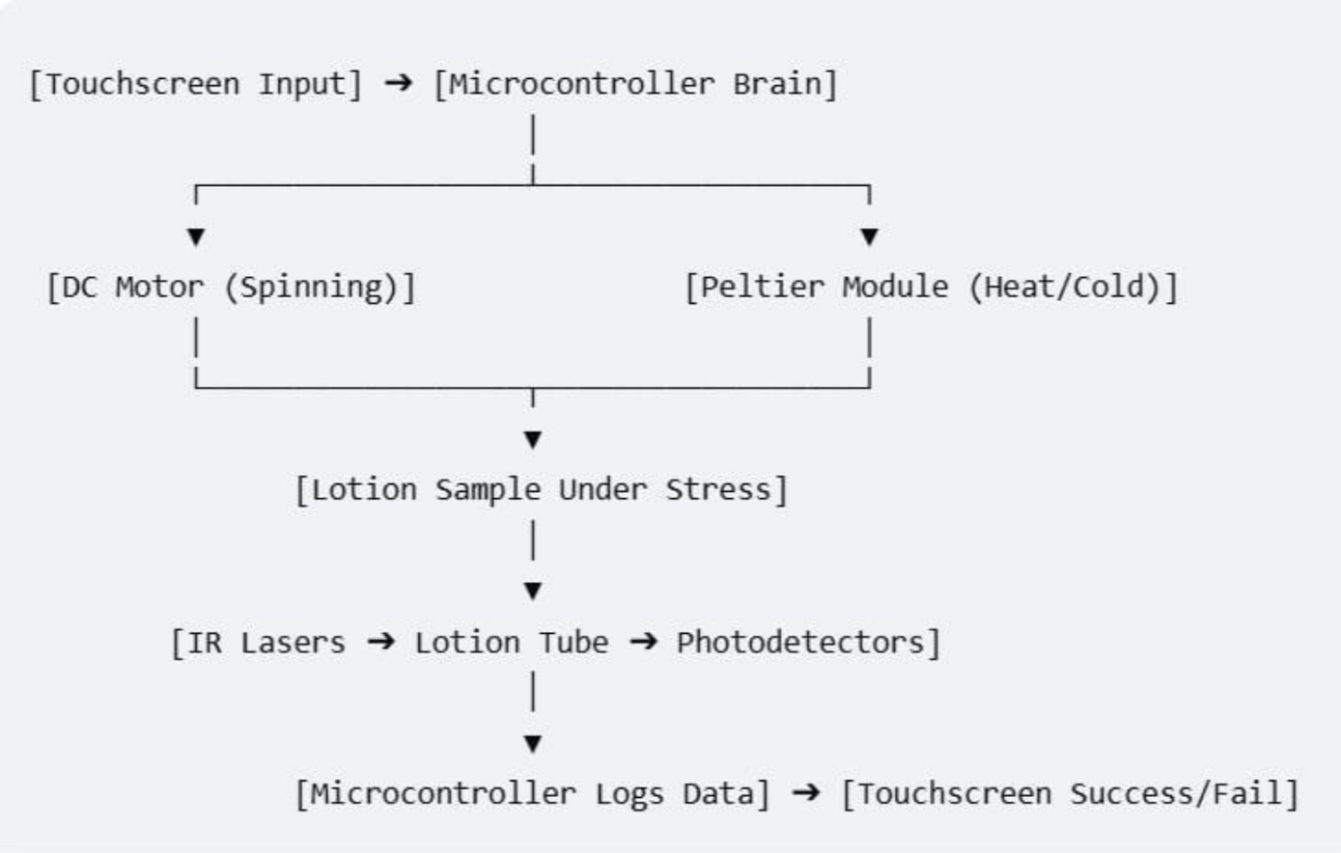
Existing Gaps & Challenges

- [1. In usual methods, they fix the temperature of the incubator and check the formulation which doesn't explain whether the product is stable in room temperature or not.]
- [2.The main gap that top brands like Nivea and Lakme face today is that due to the long time the lotion takes to get set for testing face it leads to the development of microbes and pathogens which when chemically reacted gives undesired results which the company doesn't want. So, this is where our device comes into the picture to reduce the time it takes. Because of that, growth of unnecessary microbes and pathogens will be eliminated and the desired output will be received.]
- [The main challenge that remains unsolved is time, still none of the companies are able to overcome time management. Still they're lagging behind by using manual checking for three to six months which creates a huge lag in the industry.]

Our Proposed Solution

- [Core:Core idea : Our core concept is a rapid, temperature- control stability-screening device for finished lotion. After the lotion passes it's usual initial checks, a Peltier- controlled chamber exposes it to a defined temperature profile while an ESP-32 records temperature and Photo diode readings, aiming to detect science of instability in 1-2 weeks instead of waiting 3-6 months. Our device goal is to provide early stability prediction.]
- [How the solution works at a high level:Finished lotion passes its initial checks $\rightarrow$ baseline reading $\rightarrow$ controlled chamber test $\rightarrow$ repeated IR and temperature readings $\rightarrow$ comparison with set limits $\rightarrow$ early warning if a measured change appears]
- [Main value delivered to users:The device is a screening aid; its readings must be validated against longer-term tests before they can predict shelf life.]

Technical Approach



[Explain how the system components work together::Finished lotion passes its initial checks → baseline reading → controlled chamber test → repeated IR and temperature readings → comparison with set limits → early warning if a measured change appears]

Innovation & Uniqueness

- [What is new? - Uses to predict the long-term stability of cosmetic emulsions. Aims to reduce the testing period from 3–6 months to a few hours. Analyzes formulation ingredients and their properties to identify potential stability problems earlier. Reduces the need for repeated physical experiments during the initial screening stage.]
- [Why is this approach suited to the problem? - Traditional stability testing requires a long waiting period. the device can analyze available experimental data to identify patterns associated with stable and unstable formulations. Early predictions help researchers identify promising formulations before conducting lengthy laboratory tests. It can save development time, reduce material wastage, and support faster product development]
- [What is the key differentiator? - Speed: Provides rapid predictions instead of relying only on long-term testing. Predictive capability: Estimates potential stability issues before the full testing period is complete. Reduced wastage: Helps prioritize promising formulations for physical testing. Scalability: The system could be extended to different cosmetic formulations as more reliable data becomes available.]

Feasibility & Scalability

- **Feasibility: The AL^3 360 project is highly feasible for internal RD but faces strict regulatory limitations. Financially, the high upfront equipment cost is easily justified by the rapid return on investment gained from cutting product development timelines by months. Consequently, the project is most feasible when deployed as a high-speed pre-screening tool to instantly catch formulation flaws before submitting the final candidate to traditional regulatory testing.**
- **Scalability:Scalability The scalability of the AL^3 360 project is exceptionally high because its predictive software and data models can be cloned instantly and expanded into the pharmaceutical, food, and paint industries with minimal recalibration. However, physical scalability faces standard bottlenecks, including the high capital expenditure required to replicate specialized optical hardware and the need to build distinct calibration datasets for highly diverse formulation types, such as transitioning from heavy water-based creams to oil-free anhydrous serums.**
- **Risks: The AL^3 360 project faces four primary constraints—regulatory**

Cost for Product (Optional)

- Development or hardware cost: [Cost area Estimated cost Core hardware parts =10,000–20,500 ESP32 firmware, touchscreen interface, logging, and control= 15,000–35,000 if outsourced Electronics integration, driver/safety circuit, and PCB work =10,000–25,000 if outsourced Enclosure/mechanical design and prototype fabrication labor= 8,000–20,000 if outsourced Bench debugging and basic calibration setup=5,000–15,000 Estimated total for one professionally developed prototype =48,000–1,15,500
  ]
- Hosting or maintenance cost: About 1,000–3,000/year
- Estimated total: 48,000 - 1,15,500

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: [Use Case 1: Rapid Formulation Screening User:Cosmetic Formulation Chemist / RD Specialist Scenario: AL^3 360 detects early signs of instability in new cream formulations within 4 hours, enabling rapid testing and same-day improvements.]

- Use case 2: [Use Case 2: Supplier Replacement Validation User: Supply Chain / Procurement Manager Scenario: AL^3 360 compares the stability of original and alternative emulsifiers within a day, helping validate replacements and avoid production delays.]

- Use case 3: [Use Case 3: Testing-as-a-Service (TaaS) User: Contract Testing Lab / CRO Scenario: AL^3 360 enables affordable pre-screening of cosmetic formulations, helping brands identify unstable products and reduce costly testing failures.]
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- [Benefit to the target users:Cosmetic chemists, RD teams, and manufacturers get rapid, data-driven predictions of product stability, enabling quicker decision-making and drastically reduced testing cycles.]
- [Economic: Drastically cuts product development costs, speeds up time-to-market for new launches, and lowers resource consumption for cosmetic brands and startups Environmental: Significantly minimizes chemical and material testing waste by reducing repeated, long-term physical trial iterations]
- [Expected measurable outcome:Reduces shelf-life and emulsion stability validation time from 3–6 months to just a few hours, eliminating manual inspection bottlenecks.]

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG [9]	Target [9.5]	[INDUSTRY, INNOVATION AND INFRAS-TRUCTURE]	[By using some industrial hardware device we have implemented and relevant device to chemical industry.]
SDG [13]	Taerget [13.3]	[CLIMATE ACTION]	[By using our device we are able to identify or testify that the ingredients can sustain in any temperature.]

Individual Member Contribution

Member	Contribution or responsibility
[LATHIKA B]	[Content collection, research works done, participated actively in chart work and took great responsibility as a leader.]
[LOHITHAA.M]	[Active participation in content discussion,took care of the components segment of the project and did the ppt part.]
[LAKSHYA.A.S]	[Contributed to the data collection, actively participated in chartworks, went through research papers and made ppt based on it.]
[ASMITHA.M]	[Actively took running notes for the meeting, contributed for data collection and helped to make script.]

Yukti ID

[IC201810638]